

Practical No. 9

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user ie. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

❖ CODE

```
/c/Users/acom/Desktop/OS_CD24037
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, farthest;
    int faults = 0, hits = 0;
    int found;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        found = 0;

        // Check hit
        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        // If page fault
        if(found == 0) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < framesCount; j++) {
                int nextUse = -1;

                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        nextUse = k;
                        break;
                    }
                }

                if(nextUse == -1) {
                    pos = j;
                    break;
                }
            }
        }
    }

    printf("Percentage page hit: %.2f\n", (float)hits/n*100);
    printf("Percentage page fault: %.2f\n", (float)faults/n*100);
}
```

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GNU nano 8.7

```
}

// If page fault
if(found == 0) {

    pos = -1;
    farthest = i + 1;

    for(j = 0; j < framesCount; j++) {

        int nextUse = -1;

        for(k = i + 1; k < n; k++) {
            if(frames[j] == pages[k]) {
                nextUse = k;
                break;
            }
        }

        if(nextUse == -1) {
            pos = j;
            break;
        }

        if(nextUse > farthest) {
            farthest = nextUse;
            pos = j;
        }
    }

    if(pos == -1)
        pos = 0;

    frames[pos] = pages[i];
    faults++;
}

printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}
```

 Help
 Exit

 Write Out
 Read File

 Where Is
 Replace

 Cut
 Paste



 Type here to search



❖ OUTPUT

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ nano opr.c

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc opr.c -o opr
.
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ./opr
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Frame status: 1 -1 -1
Frame status: 1 2 -1
Frame status: 1 2 3
Frame status: 1 2 4
Frame status: 1 2 4
Frame status: 1 2 4
Frame status: 1 2 5
Frame status: 1 2 5
Frame status: 1 2 5
Frame status: 3 2 5
Frame status: 4 2 5
Frame status: 4 2 5

Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ |
```